

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary Winter Examination – 2023**

**Course: B. Tech. Branch : Civil Engineering/Civil and Infrastructure Engineering**

**Semester : FOURTH**

**Subject Code & Name: BTCVC402/BTCIC402 Environmental Engineering**

**Max Marks: 60**

**Date:18-01-2024**

**Duration: 3 Hr.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

|                                                                         |     |           |
|-------------------------------------------------------------------------|-----|-----------|
| <b>Q. 1 Solve Any Two of the following.</b>                             |     | <b>12</b> |
| A) Write a note on physical characteristics of water.                   | CO1 | 6         |
| B) What is demand? Mention factors affecting demand.                    | CO1 | 6         |
| C) Give Indian Standards for quality of potable water.                  | CO1 | 6         |
| <b>Q.2 Solve Any Two of the following.</b>                              |     | <b>12</b> |
| A) What is aeration? List different types of aerators with neat sketch. | CO1 | 6         |
| B) Write a note on coagulation and different types of coagulants used.  | CO2 | 6         |
| C) Describe Lime Soda method of softening of water                      | CO1 | 6         |
| <b>Q. 3 Solve Any Two of the following.</b>                             |     | <b>12</b> |
| A) Write a short note on (i) Dead end system and (ii) Grid iron system  | CO1 | 6         |
| B) Differentiate between gravity system and pumping system.             | CO1 | 6         |
| C) State different layouts of water distribution system                 | CO1 | 6         |
| <b>Q.4 Solve Any Two of the following.</b>                              |     | <b>12</b> |
| A) State characteristic of municipal waste water.                       | CO3 | 6         |
| B) Differentiate between Municipal and Industrial waste water           | CO3 | 6         |
| C) Write a note on Composting.                                          | CO4 | 6         |
| <b>Q. 5 Solve Any Two of the following.</b>                             |     | <b>12</b> |
| A) Describe sources and effects of air pollution                        | CO2 | 6         |
| B) List different types of plume behavior with a neat sketch.           | CO2 | 6         |
| C) Write a short note on control measures of air pollution              | CO3 | 6         |

**\*\*\* End \*\*\***